RNA Torsions and Biases

Backbone and glycosidic torsions, their dynamics, and how they reveal model bias

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October 28, 2026

Background: what is a torsion?

Definition

- Four bonded atoms A–B–C–D.
- ϕ = rotation about the bond B–C.
- $\phi \in (-180^\circ, 180^\circ]$; 0° = cis, 180° = trans.

Why it matters

- Bond lengths and angles are nearly fixed, so torsions *are* the RNA conformation.
- Torsion distributions fingerprint force-field and solvent-model bias.

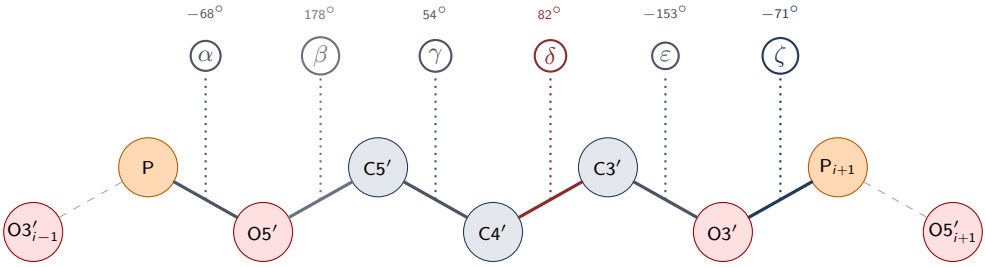
α	$O3'_{i-1}$–P–O5′–C5′
β	P–O5′–C5′–C4′
γ	O5′–C5′–C4′–C3′
δ	C5′–C4′–C3′–O3′
ε	C4′–C3′–O3′–P_{i+1}
ζ	C3′–O3′–P_{i+1}–$O5'_{i+1}$
χ	O4′–C1′–N9–C4 (purine)
	O4′–C1′–N1–C2 (pyrimidine)

What each torsion does

	A-RNA	Function
α	-68°	Backbone twist at P–O5′; flips in loops and non-canonical motifs
β	178°	Nearly always trans; rigid link into C5′
γ	54°	Sets the C5′–O5′ direction ($g^+/t/g^-$) relative to the sugar
δ	82°	**Sugar pucker**: $\sim 80^\circ$ = C3′-endo (A-form), $\sim 145^\circ$ = C2′-endo
ε	-153°	Sugar-to-phosphate link; coupled to δ and ζ
ζ	-71°	Pairs with ε to set backbone kinks and turns
χ	-158°	**Base orientation**: anti (A-form) vs. syn ($\sim +60^\circ$)

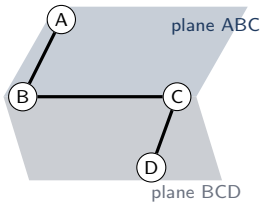
Takeaway: δ and χ are the most informative reporters.

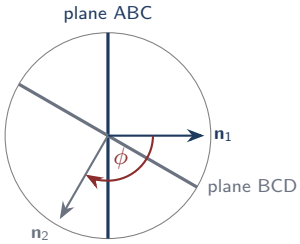
The RNA backbone and its six torsions



Each torsion is the rotation about its highlighted bond. Values: A-RNA reference.

Planar view and the normal-vector definition





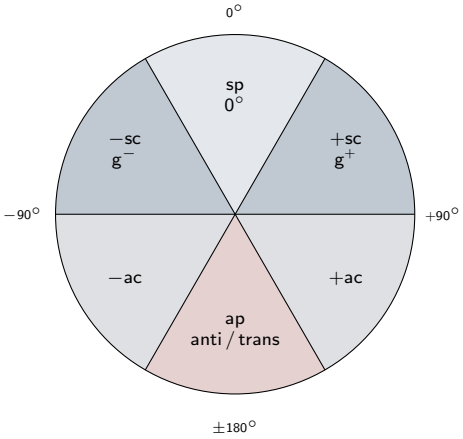
view along $\mathbf{b}_2$: planes edge-on

$\mathbf{b}_1 = \mathbf{r}_B - \mathbf{r}_A,\ \mathbf{b}_2 = \mathbf{r}_C - \mathbf{r}_B,\ \mathbf{b}_3 = \mathbf{r}_D - \mathbf{r}_C \qquad \mathbf{n}_1 = \mathbf{b}_1 \times \mathbf{b}_2,\ \mathbf{n}_2 = \mathbf{b}_2 \times \mathbf{b}_3$

$$\phi = \operatorname{arctan2}\big(\, y,\ x \,\big), \qquad x = \mathbf{n}_1 \cdot \mathbf{n}_2 \ \propto \cos\phi, \qquad y = |\mathbf{b}_2|\, \mathbf{b}_1 \cdot \mathbf{n}_2 \ \propto \sin\phi$$

$\arctan 2(y, x)$ puts the point (x, y) in the correct quadrant, giving the full -180° to 180° range and the sign.

Torsion ranges: gauche, anti, syn



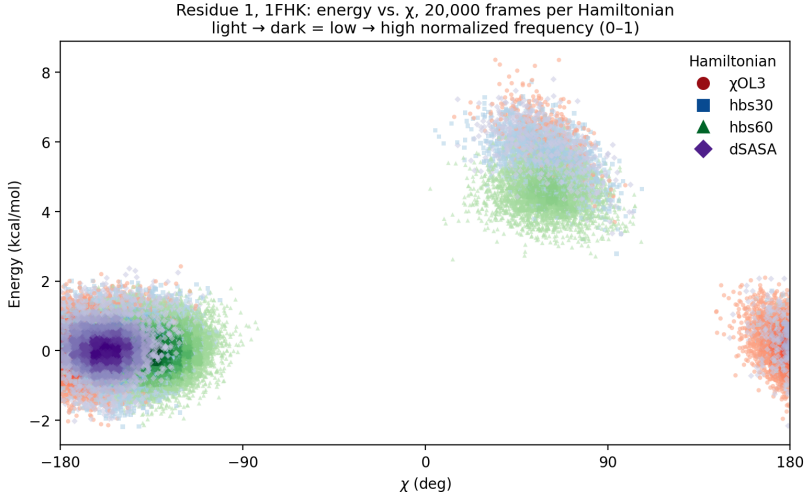
sp = synperiplanar (cis)
sc = synclinal = gauche ($\pm 60^\circ$)
ac = anticlinal ($\pm 120^\circ$)
ap = antiperiplanar = anti/trans (180°)

χ: **anti** $\approx -160^\circ$ (A-form), **syn** $\approx +60^\circ$

A-RNA reference

α	β	γ	δ	ε	ζ	χ
-68	178	54	82	-153	-71	-158

Energy vs. χ: residue 1 of 1FHK



Schematic, synthetic data